Beyond the Original Sin: Lay Evaluations of Creative Contribution in AI-Assisted Artworks

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Abstract

United States copyright protects only a person's contribution to a work built on AI-made material, so whoever applies it must judge the human part while knowing that the base was machine-made. We asked whether lay evaluators can. In five studies, 747 students rated AI-made images that a label attributed to an artist's design work or to his prompting of an AI system. On the same images, the AI label lowered creativity ratings by 0.45 points and effort ratings by 0.92 on 1-to-5 scales (Study 1); stating that the artist had spent up to 100 hours did not close the gap. When he then edited each image by hand, ratings followed how much he changed it, and, compared across studies, the label carried much less into them: small and uneven, but not zero. An AI label acted more like information about who did what than like a stain on the work.

Introduction

The US Copyright Office, which examines and records copyright claims, can register a person's contribution to a work made with artificial intelligence (AI), such as the person's own text, a creative selection and arrangement of AI material, or modifications original enough to be protected.[1,2] Originality means independent creation with at least a minimal degree of creativity.[3] AI material beyond a trivial amount must be disclaimed, that is, excluded from the claim.[1,4] The comic book *Zarya of the Dawn* is registered for its text and its selection and arrangement of text and images, not its AI images.[5] The Office takes this limit from section 103(b) of the Copyright Act: copyright in a compilation, such as *Zarya*'s arrangement, or in a derivative work, one based on an earlier work, covers only what its author, the person who created it, contributed.[1,6,7] It treats edited AI images as derivative works only by analogy, because in its view wholly AI-made images are not works of authorship.[2] Copyright requires a human author: in *Thaler v. Perlmutter*, which concerned a machine named as sole author, a federal appeals court held that the Act requires one.[2,8] In 2025 the Office concluded that, with the technology then available, prompts alone gave a user too little control to author the output.[1,2] These rules rest on no claim about how people judge AI-made work, yet they ask whoever applies them to judge a person's part on its own while knowing the base was machine-made: examiners, who

read applicants' disclosures of AI content,[1] and jurors, because originality can go to a jury in an infringement suit.[9]

We call original sin the hypothesis that lay judgment does not separate the two parts: that an AI origin acts like a stain passing to whatever is built from it. It concerns people, not law; our debriefing wrongly told participants that a categorical version was law (Methods). It extends to AI the belief, studied as contagion, that a person's essence passes into objects they touched or merely designed.[10,11] The same image is rated lower when labeled AI-made than human-made,[12–14] a drop we call the label's penalty, small to moderate in a meta-analysis[15] and not found in every design.[16,17] Work described as made by a person with AI is also rated lower on most measures, even a painting made by hand from AI-supplied compositions, though its artist was rated more creative.[13,18] These studies rated the finished work or its maker. Apart from our earlier report of two of the studies below,[19] we know of no study that showed an image beside a person's edit of it, varied only its described origin, and asked about the resulting second work.

On the effort-signal account, the label lowers ratings of originals only because it signals little work, as AI-labeled work is judged to take less effort and time;[12,13,20] a stated 100 hours should then shrink the penalty to almost nothing. If the label instead says the act was prompting, not drawing, the penalty should persist. Because copyright rewards "originality, not effort,"[3] we ask whether the label lowers effort ratings more than creativity ratings. For second works, Study 2's registration (a dated record of its hypotheses and analyses, not a copyright registration) sets out three accounts, which we call taint, dilution and severance. Taint is original sin applied to second works: the same penalty at every level of modification, at full strength as large as on originals. Under dilution, raters judge the second work partly by its original, so the penalty shrinks as the edit grows. Under severance, raters judge only what the person added, so there is no penalty, which parallels section 103(b). After seeing the results, we stated a principle, scoped credit: people's ratings credit a person for what they take that person to have done, and a label says which act produced what they see. It implies a penalty on originals, whose making the label describes; little carry-over to second works, whose making both groups read about alike; and an effect of stated hours only when they describe the rated act.

A label told each participant either that the artist, Alex, had made the originals with design and sketching tools or that he had prompted an AI system to make them; all were in fact made with OpenAI's DALL·E through ChatGPT, so any group difference is a label effect. Studies 1 and 3 asked about the originals, Study 3 stating a time of under 1, 10 or 100 hours. In Studies 2, 4 and 5, both groups read, in the same words, that Alex made three second works from each original with his design and sketching tools: None, the original unchanged; Slight, a light edit; and Dramatic, a heavy one. We made the edits by hand. Study 4 also stated hours for each original, and Study 5 hours for each edit. Participants saw each second work beside its original before rating it. The questions concerned the new work and the edit, not how the original was made, so any label effect is a carry-over; its absence would parallel section 103(b) but would also follow if raters answered the question asked (Discussion). All five studies were registered on 8 October 2026, after data collection, so none was preregistered; only Study 5 was registered before any of its ratings had been averaged or compared by label. The numbering follows the argument; Studies 2 and 4 ran first (Methods).

Results

Fig. 1 and Table 1 summarize the five studies. By credit we mean the ratings people gave a work, not judgments of authorship. The 771 analyzed records are participations, not people: they came from 747 students in the UCLA Psychology subject pool, 24 of whom took part in two studies, all for course credit.

Supplementary Table 1 lists every registered test with its status and verdict. A p-value is the chance of a difference at least this large if there were no effect, and a result is significant when p is below .05. A 95% confidence interval (CI) is the range of values the data do not rule out at the 95% level. A registered confirmatory test is decided by its registration's rule, usually after Holm's correction,[21] which raises each p-value in a set of tests so that the chance of calling any result in the set significant by mistake stays at .05. A registered secondary test lies outside that set and is uncorrected unless stated. A registered description is an estimate that a registration asks us to report without a test of its own, and a descriptive result carries no test. An exploratory analysis was named as exploratory in a registration or not named at all; the analyses that pool Studies 2, 4 and 5, the image check and the breakdown of Study 5's interaction were specified on 9 and 10 October 2026, after the registered results were known. Because every registration followed data collection, and for Studies 1 to 4 followed analyses of the same data (Table 1), 'registered' means written into a registration, not fixed before collection.

An AI label lowers credit for the same image, more for effort than creativity

Study 1's 111 participants rated all ten originals on 1-to-5 scales for creativity ('How much creativity was involved in the production of this work? Creativity is defined as the presence of at least a minimal original, intellectual contribution to the creation of the work.') and for effort ('How much effort was involved in the production of this work?', with no definition). Each estimate is a difference between predicted mean ratings from a linear mixed model, a regression that gives each participant and each image its own baseline. We call the Human-label minus AI-label difference the label's penalty; a positive penalty means that the AI label lowered the rating. With no unlabeled condition, 'lowered' means lower than under the Human label.

The AI label lowered both ratings of the same images: by 0.45 points on creativity (95% CI 0.19 to 0.70; Holm $p = .001$; Fig. 2a) and by 0.92 points on effort (0.62 to 1.22; Holm $p < .001$). The effort penalty was larger by 0.47 points (0.28 to 0.67; Holm $p < .001$; Fig. 2a, gray row), and an ordinal model, which treats the five answers as ordered categories rather than evenly spaced numbers, agreed ($p < .001$). Under the Human label, effort sat above creativity (3.76 against 3.55); under the AI label, it sat below (2.84 against 3.10). These results pass Study 1's registered confirmatory tests of H1, that the AI label lowers both ratings, and H2, that it lowers effort more. But both drops had been seen in t-tests (comparisons of two means) in April 2025, and the larger drop on effort in a talk in September 2025, before the model and correction were chosen, so H1 and H2 are registered tests of patterns already seen, not predictions. Only the creativity item carried a definition, so part of the H2 difference may come from the wording of the items. Study 1's H3, that the penalty is the same for images with and without a human figure, was decided outside the confirmatory set by its own registered rule, against a bound set

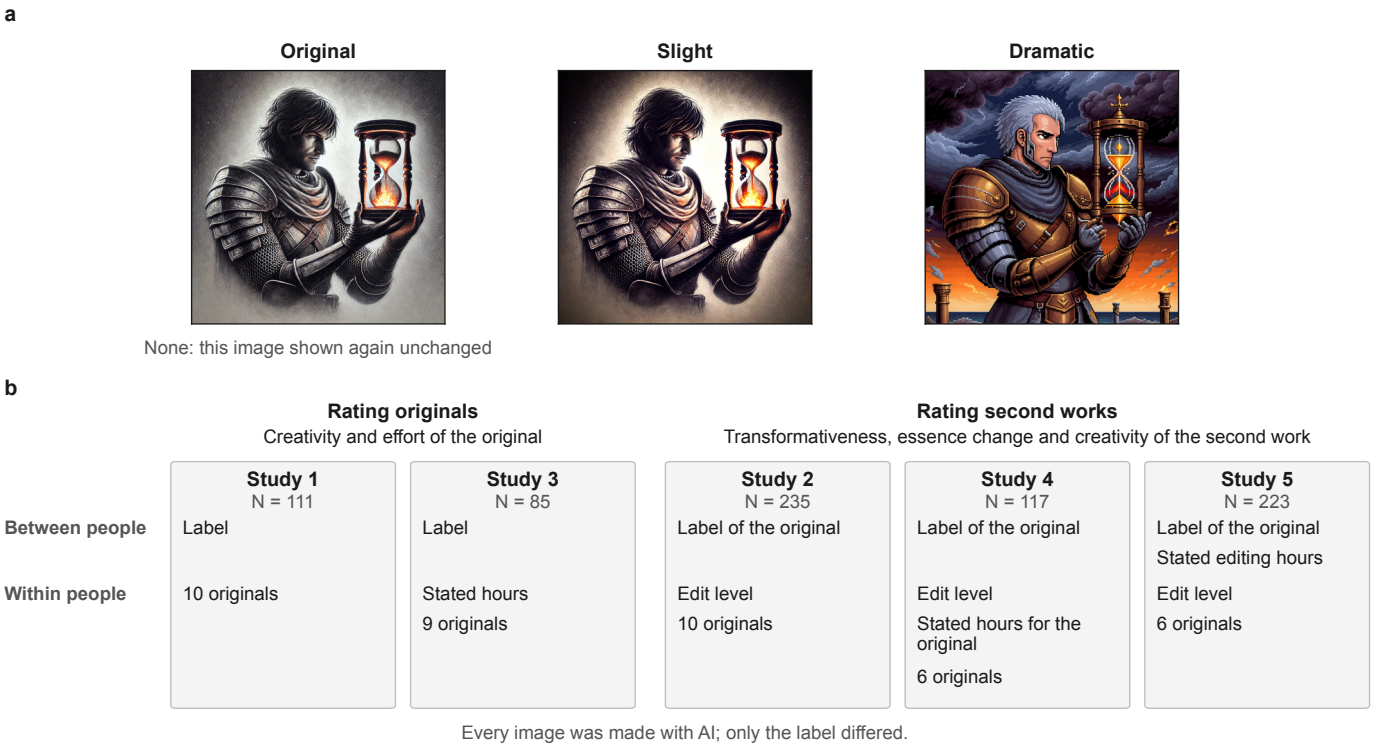


Figure 1: Design of the five studies. **a**, One of the ten originals and its Slight and Dramatic versions, which we edited by hand. In Studies 2, 4 and 5 the original was also shown again unchanged, as the None second work, so each second work's edit level was None, Slight or Dramatic, and each second work appeared beside its original. Of the ten Slight versions, 7 were also cropped or resized; the one shown was not. **b**, What each study varied between participants (each person saw one condition) and within participants (each person saw every condition), what participants rated and how many were analyzed (N). The label said that the artist, Alex, made the original with design and sketching tools (Human label) or by prompting an AI system (AI label). Every original was made with OpenAI's DALL·E, used through ChatGPT, so the Human label was a cover story, and each participant saw one label throughout. In Studies 2, 4 and 5, both label groups read that Alex made the second works with his design and sketching tools, so the label described only the original. Stated times were under 1 hour, 10 hours or 100 hours; Study 3's were hours of drawing under the Human label and of revising prompts under the AI label. Study 5 also stated a time for the original, but a fault in the survey script left its record blank, so it is not analyzed. The studies ran in the order 2, 4, 1, 3, 5, and all five were registered after data collection: their hypotheses and analyses were filed on the Open Science Framework, an online archive for study plans, which has nothing to do with copyright registration.

Table 1: The five studies at a glance. Supplementary Table 1 lists every registered test and its verdict; Supplementary Table 2 gives the label's effect on second works, with intervals.

Study	What it asked	Registered answer	Order run; registered
Originals: the same AI-made images, rated for creativity and effort			
Study 1 $N = 111$	Does the AI label lower both ratings (H1), effort more (H2), alike with or without a human figure (H3), and spread them more between people (H4)?	H1 and H2: supported. H3: supported on effort, inconclusive on creativity. H4: not supported as registered.	Ran third; registered after its analyses
Study 3 $N = 85$	With under 1, 10 or 100 hours stated, does the gap (the penalty) close by 100 hours?	Decision on both ratings: the gap persists. The AI label lowered ratings and stated time raised them (both supported); a change in the gap with time was not significant.	Ran fourth; registered after an Apr. 2025 analysis and a Sept. 2025 talk
Second works: Alex's edits of each labeled original, at three levels			
Study 2 $N = 235$	Do ratings of a second work rise with the level of edit (H1)? Is the label's effect small (H2), and does it shrink from None to Dramatic (H3)?	H1 holds on every measure. H2, over the three measures: not shown to be small (small or absent on transformativeness and essence change; not shown to be small on creativity). H3: not significant.	Ran first; registered after a Feb. 2025 analysis[19] and a Sept. 2025 talk
Study 4 $N = 117$	Study 2's H1 and H2 again, and do hours stated for the original change ratings of the second work (H3)?	H1 supported. H2, over the three measures: not shown inside the bound (inside it on essence change only, a pass lost in the registered refit[a]). H3: not significant.	Ran second; registered after a Feb. 2025 analysis[19] and a Sept. 2025 talk
Study 5 $N = 223$	Do hours stated for the edit change ratings of the second work, or the label's effect on them?	Editing hours: absent or small, inside ± 0.219. A change in the label's effect with editing hours: significant on creativity only. The label's effect (secondary): significant, not inside the bound, at Slight on every measure and at Dramatic on transformativeness.	Ran last; registered before any rating was averaged or compared by label

Every original was AI-made (OpenAI's DALL·E, used through ChatGPT). Human label: Alex used design and sketching tools; AI label: he prompted an AI system. *Penalty*: Human-label minus AI-label mean rating on 1-to-5 scales (the gap, in Study 3's registration). *None, Slight, Dramatic*: the original shown again unchanged, a light edit, a heavy edit; the measures: transformativeness, essence change, creativity. *Supported*, *holds*: passed by the registration's rule. *Significant*: p below .05, after Holm's correction for confirmatory tests (which holds the chance of any false positive in a study's confirmatory set at .05) and uncorrected for *secondary* tests outside that set. *Small or absent*, *inside the bound*: an equivalence test placed the effect inside ± 0.224 points, half of Study 1's creativity penalty (± 0.219, Study 3's creativity time effect, for editing hours); small, not zero. *Not shown to be small*, *not shown inside*: that test failed, which does not show an effect. *Inconclusive*: no effect and a small one could not be told apart. [a]Without the 7 Human-label participants who wrote that a machine made the work. All five registrations were filed on 8 October 2026, after data collection.

after the analyses. It was supported on effort and inconclusive on creativity, where the design could not tell no difference from a small one. H4, a registered confirmatory test that the AI label spreads ratings more between people and less within each person, met its criterion on effort only, so it was not supported as registered (Supplementary Note 1 and Supplementary Fig. 1).

A registered robustness check (descriptive) refit the penalty 18 ways for each rating, and in all 36 versions the AI label still lowered the rating. Of the 56 Human-label participants, 25 wrote at least once that a work is or looks machine-made, by a word list written after results by label had been read; this is a count of statements, not of disbelief. Disbelief in the cover story would shrink the penalty, not create it, and a registered refit without those participants enlarged it, to 0.55 points on creativity and 1.17 on effort. Dropping low raters can enlarge a penalty by arithmetic alone. Random Human-label groups of the same size and mean effort rating enlarged the creativity penalty, measured against the spread of ratings, as much as the exclusion did, but never enlarged the effort penalty that far; so on effort the enlargement is not explained by that arithmetic, and these data say nothing more about its cause (exploratory description; Supplementary Note 2).

Stated hours raised ratings but did not close the gap within 100 hours

Study 3 tested a rival account named in its registration, which we call the effort-signal account. On it, the AI label lowers ratings only because it signals little work, so stating that Alex spent 100 hours on an image should shrink the penalty (the registration's gap) to almost nothing. Study 3's 85 new participants each rated nine originals, three at each stated time: under 1 hour, 10 hours or 100 hours. Under the Human label Alex spent those hours drawing; under the AI label he spent them revising prompts, so there more time also meant more control over the image, and the two cannot be told apart. The registered rule called the gap persistent if its 95% CI at 100 hours lay above zero and it failed an equivalence test, which asks whether an effect is shown to lie inside a stated bound, here ± 0.224 points (half of Study 1's creativity penalty).[22,23]

On both ratings, the registered decision was that the gap persists. At 100 hours, the gap's 90% CI, the interval that matches an equivalence test, ran from 0.11 to 0.62 points on creativity and from 0.20 to 0.78 on effort, and both 95% CIs lay above zero. This does not refute the effort-signal account: both 90% CIs reach below 0.224, so part of the gap may still come from how little work the AI label implies, especially on effort.

The registered confirmatory F tests for label and for time, which ask whether a factor changes the mean rating, survived Holm's correction on both ratings; by the estimates, the AI label lowered both ratings and stated time raised both. Averaged over labels, 100 hours minus under 1 hour was 0.22 points on creativity (95% CI 0.07 to 0.36), most of it between 10 and 100 hours, and 0.75 on effort (0.60 to 0.90). A change in the gap with time (the label-by-time interaction) did not survive Holm's correction on either rating, and no test of equal rises under the two labels was registered. As registered description, the creativity gap was 0.35, 0.35 and 0.36 points at under 1, 10 and 100 hours, and the effort gap was 0.58, 0.26 and 0.49 (Fig. 2a). Each effort gap was smaller than Study 1's effort penalty of 0.92, an untested comparison across studies. With time stated, the two gaps were close (at 10 hours the effort gap was the smaller), and Study 3 registered no test of effort against creativity, so the larger drop on effort rests on

Study 1 (Supplementary Note 3).

Ratings of Alex's second work follow the level of modification

Studies 2, 4 and 5 changed two things. The rated act became Alex's edit, described in the same words to both label groups, and his part became visible as a before-and-after pair. Each original appeared beside three second works: None, the original shown again unchanged; Slight, a light edit; and Dramatic, a heavy one. Participants rated each second work's transformativeness, how much it adds new expression, meaning or message to the original (a lay rating whose wording echoes court decisions on fair use, the rule that allows some copying without permission; it is not a fair-use judgment). They also rated its essence change, how much the original's essence changed (defined in Studies 4 and 5 as its core sense and feeling), and the creativity involved in its production. The label now described an act the questions did not ask about, so any effect it had is a carry-over from what was said about the original. Three accounts predict different carry-overs. Under taint, the AI origin passes to anything built from it, so the penalty is the same at every level and, at full strength, as large as on originals. Under dilution, it is largest at None and shrinks as the edit grows. Under severance, the label does not reach second works at all. These studies also differed from Studies 1 and 3 in participants, timing, label wording, how many originals each person saw (six of the ten in Studies 4 and 5) and measures (effort was not rated, and Study 2's creativity item carried no definition), so every comparison between originals and second works is a reading across studies, not a test.

Ratings rose steeply with the level of modification (H1 in Studies 2 and 4, registered confirmatory; creativity in Fig. 2b). In both studies every step was positive and significant after Holm's correction on all three measures (every Holm $p < .001$), by 0.70 to 0.99 points from None to Slight and 1.41 to 1.73 from Slight to Dramatic. The ordinal model agreed, except on Study 2's creativity ratings, where its fit did not converge (its fitting procedure settled on no solution), so that check is not completed. Study 5's predicted means rose the same way (descriptive). In an exploratory check in which two image-recognition networks measured each edited image's distance from its original,[24,25] every Slight version lay closer to its original than the Dramatic version did, but within a level, ratings are not shown to track the amount of visible change (descriptive; Supplementary Note 4). At None, means in Studies 2 and 4 sat near the floor of the scale (1.01 to 1.22), leaving little room for a label effect. Because the label varied between participants and the level within them, the sizes of their effects cannot be compared.

The AI label's effect on the second work is much less than on originals, but not nothing

Studies 2, 4 and 5 asked whether the label's penalty on a second work lay inside a registered bound of ± 0.224 points, half of Study 1's creativity penalty on originals (0.448 points before rounding; Fig. 2b; Supplementary Table 2). Judged alone, an equivalence test passes when the penalty's 90% CI lies wholly inside the bound; inside a Holm-corrected set, it can fail even then. A pass means that the penalty is small, not zero; a failure means only that it is not shown to be small.

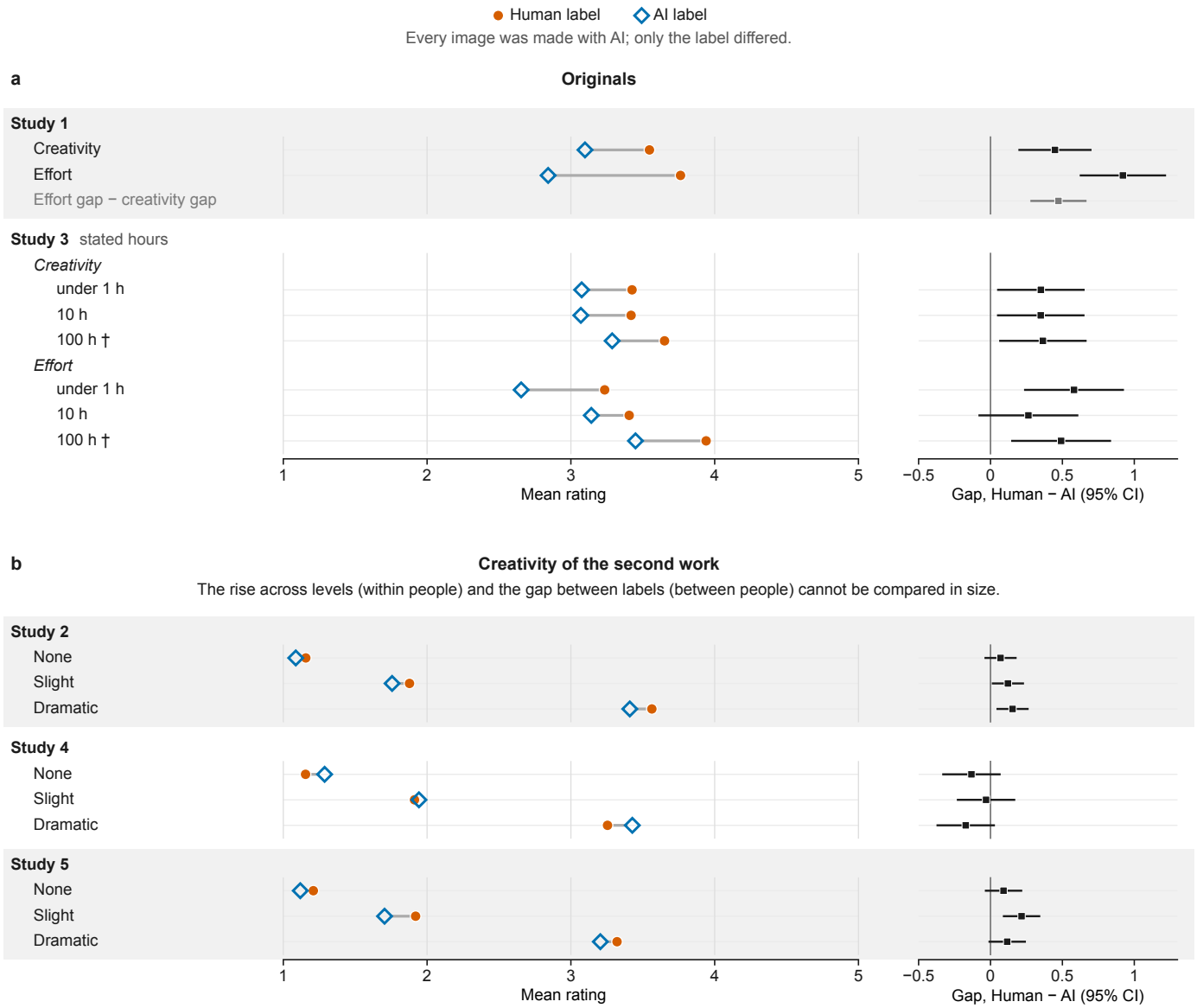


Figure 2: Mean ratings under the two labels, and the gap between them. Circles and diamonds are the two label groups' mean ratings (1 to 5), as predicted by each study's statistical model (Methods). Squares are the gap, the Human-label mean minus the AI-label mean (called the penalty in the text), at the same scale as the ratings, with its 95% confidence interval (CI). Right of zero, the AI label lowered the rating. Judge a gap by its CI, not by whether the marks overlap; a CI that crosses zero does not show a gap of zero. 'Registered' means written into a registration filed after data collection, not copyright registration. **a**, Originals. Study 1's gaps are registered confirmatory tests of H1, that the AI label lowers both ratings, and its gray row tests H2, that it lowers effort more; all three passed after Holm's correction. Study 3's gaps are registered description, estimates its registration names with no test of their own. The 100-hour gaps (†) entered the registered decision, which was that the gap persists. **b**, Creativity of the second work, asked as the creativity involved in its production; None is the original shown again unchanged. Each person saw every level but only one label. The gaps are exploratory in Study 2 and registered secondary tests in Studies 4 and 5, outside each study's confirmatory set (Supplementary Table 1) and judged at .05 without correction. They average over the hours stated for the original (Study 4) or over stated editing hours (Study 5, whose creativity gap differed across them; Supplementary Fig. 2). Supplementary Fig. 3 shows the other two ratings; Supplementary Table 2 lists the registered tests. No test compared rows from different studies.

Averaged over levels, the penalty was shown inside the bound in 3 of the 6 registered confirmatory tests (Study 2's and Study 4's H2): transformativeness and essence change in Study 2, and essence change in Study 4. H2 asked for a small effect on every measure, so as registered it holds in neither study. Study 2's creativity penalty, 0.11 (90% CI 0.03 to 0.20), was not shown to be small: its uncorrected interval lay inside the bound, but the test failed after Holm's correction (Holm $p = .068$). That penalty had already appeared in our published analysis of these data,[19] run by February 2025, in which this study is Study 1, so this test re-analyzes an effect already known. Study 4's transformativeness test also failed only after correction (Holm $p = .064$), and its creativity estimate ran the other way (-0.11; 90% CI -0.27 to 0.04), not significantly. In a registered refit without the 7 Human-label participants who wrote that a machine made the work, Study 4's essence-change pass was lost (Holm $p = .020$ on the full sample, Holm $p = .0501$ in the refit), the only one of the refit's 15 verdicts to change.

Study 5 registered the penalty at each level as secondary tests, judged without correction. At None, where ratings sit near the floor, the penalty lay inside the bound on all three measures. At Slight, it was significant and not shown inside the bound on all three: 0.13 points on transformativeness ($p = .049$), 0.17 on essence change ($p = .006$) and 0.22 on creativity ($p = .001$). Each of these estimates is under 0.224, so 'not inside the bound' means not shown to be small, not shown to be large. They also average over the three stated editing times, and in an exploratory breakdown the Slight creativity penalty rests largely on the 10-hour groups (next section; Supplementary Note 5 and Supplementary Fig. 2). At Dramatic, transformativeness was also significant and not inside the bound (0.17), essence change lay inside it, and creativity was inconclusive (0.12; its 90% CI reached 0.225, just past the bound). Study 4 registered the same tests at each level as secondary tests. At Slight they, too, found penalties that were significant and not shown inside the bound on transformativeness and essence change, while the creativity penalty lay inside the bound (Fig. 2b and Supplementary Fig. 3; Supplementary Note 6).

Across the 24 registered equivalence tests on second works, every point estimate was smaller than 0.224 in size, but only 11 tests showed the penalty inside the bound. Read across studies rather than tested, every registered interval for the creativity penalty on second works excludes Study 1's full penalty of 0.448 points, the size that full-strength taint predicts, although several overlap that penalty's own 95% CI (0.19 to 0.70). On this reading, the label carried much less into creativity ratings of second works than into creativity ratings of originals; creativity is the only measure rated on both. What it did carry, which we call the residue, was small on average, uneven across studies and detectable at slight edits in Study 5 on every measure and in Study 4 on transformativeness and essence change; in Study 5 the exploratory breakdown places most of it in the 10-hour groups.

No detectable carry-over from hours stated for the original, and no detectable effect of stated editing hours on average

Study 4 stated how long Alex had spent on each original, and each participant saw every stated time. The hours did not significantly change ratings of the second work on any measure, alone or by level of modification (Study 4's H3, confirmatory; every Holm $p > .999$), and the ordinal

model agreed. As registered secondary tests, 100 hours minus under 1 hour lay inside ±0.224 points on all three measures (transformativeness −0.02, essence change 0.01, creativity 0.03), and no change in the label's penalty across those hours was detected (not significant). Our published analysis of these data,[19] in which this study is Study 2, reported an effect of these hours on creativity that the registered model does not reproduce.

Study 5 stated how long Alex had spent making the second work, one time per participant for the whole session. Averaged over the two labels, stated editing hours did not detectably change ratings (confirmatory; every Holm $p > .999$). Against a bound of ±0.219 points, the size of Study 3's creativity time effect, the registered secondary equivalence tests (uncorrected) read 'absent or small: inside the bound' on all three measures. On creativity, 100 hours minus under 1 hour was 0.04 points (90% CI −0.08 to 0.15), so effects up to about 0.15 remain compatible. Neither Study 4 nor Study 5 rated effort, on which Study 3's time effect was largest.

Only 1 of Study 5's 6 registered confirmatory tests passed: the label's creativity penalty differed across stated editing times ($F(2, 216.8) = 4.93$, Holm $p = .048$), and the registered ordinal check and a reanalysis with every item rescaled from 0 to 1 agreed. Rather than shrinking as the hours rose, as the registration had sketched, the penalty in an exploratory, uncorrected breakdown (Supplementary Note 5 and Supplementary Fig. 2) was small at under 1 hour and largest at 10 hours, where it reached 0.48 at Slight and 0.49 at Dramatic, about as large as on originals, read across studies. At 100 hours it reversed at Dramatic (−0.25). Each label-by-time combination was a separate group of 35 to 38 people, and the 10-hour groups also showed the largest penalties on the other two measures, whose registered interactions were not significant. A single 10-hour group that happened to rate unusually could produce this pattern on all three measures and inflate Study 5's Slight and Dramatic estimates; with the corrected p just under .05, the interaction may also be a chance result that survived correction.

In an exploratory pooling of Studies 2, 4 and 5, the residue was largest at slight edits

Specified after every registered result was known, this analysis is uncorrected and changes no registered verdict (Supplementary Note 6; Supplementary Fig. 4). A model that fitted Studies 2, 4 and 5 together, each weighted equally, put the creativity penalty at 0.01, 0.10 and 0.03 points at None, Slight and Dramatic; its fits were singular, meaning one of the sources of variation the model allows for was estimated at zero. On all three measures the Slight penalty lay above the average of the None and Dramatic penalties, and no fall from None to Dramatic was shown, so the pattern dilution predicts did not appear, though the floor at None weakens that test. The pooled curve averages over studies that disagree on creativity: without Study 4, whose estimates lay below zero (none significantly), the Slight penalty rose to 0.17 and its 90% CI (0.10 to 0.24) extended past the bound. Two standard scores for choosing between models disagreed: AIC (Akaike's information criterion) favored a penalty that differs by level, and BIC (the Bayesian information criterion), which charges more for each added quantity a model estimates, favored no penalty at all, as severance predicts. The residue was not confined to Slight: Study 2's creativity penalty (exploratory) and Study 5's transformativeness penalty (registered secondary) were significant at Dramatic.

What the explanations referred to

This section is exploratory (Supplementary Note 7). Eight fixed word lists, applied by a script blind to label and rating, flagged what each written explanation mentioned. Because each explanation followed its rating, these results describe what the words referred to, not why anyone rated as they did. On originals, what 'effort' referred to depended on the label. In Study 1, 62.5% of Human-label effort explanations named visible detail, and 50.5% of AI-label effort explanations mentioned prompting or machines, against 15.4% under the Human label; Study 3 showed the same shift (uncorrected). On second works, the label left only a faint trace. AI-label explanations mentioned prompting or a machine more often than Human-label ones, but rarely (at most 5.9%), and no difference was shown in how far a person's explanations changed from None to Dramatic, where taint predicts less change under the AI label. No study asked participants to recall the label (a manipulation check).

Discussion

Asked about an image that a label tied to an AI system, people gave it less credit; asked about a person's own rework of that image, they gave its AI origin little weight. The AI label lowered ratings of the same images, more for effort in Study 1, and a stated 100 hours did not close the gap. Once Alex had edited each image by hand, ratings followed how much he had changed it, and the label carried much less into them than into ratings of the originals. That is a reading across studies, not a test, and the carry-over was not zero: it was small, uneven and detectable at slight edits in Study 5. These ratings do not support the original-sin hypothesis in its strong form. An AI label behaved more like information about who did what than like a stain on the object.

Table 2 sets the accounts named in the Introduction against this evidence. The label did not work only by signaling little work, since a stated 100 hours left the gap in place, though a partial role for that signal, above all on effort, is not excluded. Neither taint at full strength nor dilution is supported. Severance comes closest: one standard score for choosing between models favored no penalty on second works at all, and the penalties that stand against it come from uncorrected tests, most of them at slight edits. On the originals the images were identical, so the penalty came from the label, but these data cannot separate the account we favor from contagion, the belief that an object carries something of how it was made. That account, scoped credit, which we stated after seeing the results, holds that people credit a person for what they take that person to have done and read a label as saying which act produced what they see. It predicted a penalty on originals, credit that rises with the time stated for the act the label describes, and little carry-over to a different act, and each held. It also predicted that hours stated for Alex's edit would raise credit for the edit, and they did not (creativity 0.04 points, 90% CI -0.08 to 0.15). A single stated time may be hard to weigh without another to compare it with, or a visible edit may leave a stated time little to add. Both readings came after the fact; stating several editing times to the same people would separate them.

The most important limit on this answer is what the questions asked. The second-work items asked about the new work and the edit, so a rater who answered them literally had little reason

Table 2: Accounts of the AI label's effect, and what the five studies show. The penalty is the mean rating under the Human label minus the mean under the AI label. Comparisons between originals and second works are readings across studies, not tests. Scoped credit was stated after the results were known.

Account	What it predicts	What the studies show	Reading
Effort signal: the label only signals little work	A stated 100 hours closes the gap on originals	The registered decision on both ratings: the gap persists; its 90% CIs at 100 hours reach below ±0.224 (Study 3)	Not the whole story; a partial role, above all on effort, is not excluded
Taint: an AI origin passes to whatever is built from it	The originals' penalty, at full strength, at every level of edit	Every registered interval for the creativity penalty on second works excludes Study 1's 0.448 points	Not supported at full strength
Dilution: the penalty shrinks as the edit grows	Largest at None, falling toward Dramatic	Study 2's registered test was not significant and ran the other way; no fall in the pooled model (exploratory); the floor at None weakens both	Not supported
Severance: the label does not reach second works	No penalty at any level	A penalty at Slight on all three ratings in Study 5 (registered secondary, uncorrected); BIC favored no penalty in the pooled model (exploratory)	Not supported in its strict form, but close to the data
Visible traces: ratings follow what the image shows	No label effect on identical images	A penalty on identical originals in Studies 1 and 3 (registered confirmatory)	Cannot explain the originals' penalty
Contagion: an object carries something of how it was made	A penalty on originals that also reaches what is built from them	A penalty on originals; a small, uneven carry-over to second works	Not separable from scoped credit on originals; a strong version is weakened
Scoped credit: credit follows the act the label describes	A penalty on originals; stated time raises credit for the act described; little carry-over to a different act; stated hours of the edit raise credit for the edit	The first three held; stated editing hours showed no detectable effect (creativity 0.04 points, 90% CI −0.08 to 0.15; Study 5, registered)	Organizes most results; one failed prediction

to use the label. Scoped credit and a literal reading therefore predict the same small carry-over, and these studies cannot tell them apart. A closer study suggests that the difference matters. When an artist painted the same landscape by hand and viewers were told that an AI tool had supplied the compositions, ratings of the painting as a whole fell on most measures, while the artist's creativity was rated higher.[18] Together, the two sets of results suggest that an AI origin may count against a whole work but not against a person's contribution when the question isolates that contribution, a hypothesis that asking both questions about the same pairs would test. The legal questions are scoped in just this way: a registration claims only what a person contributed, with the AI material disclaimed.[1]

For the law, the ratings show parallels and divergences, not a match. The label lowered effort more than creativity in Study 1, and *Feist* set labor aside as a ground of originality while keeping a minimal creative spark;[3] but Human-label effort explanations named visible detail far more often than time or skill (exploratory), so the effort rating is no measure of labor. Stated hours raised creativity ratings of originals by 0.22 points, a departure from *Feist*'s rejection of labor, though under the AI label more time also meant more control over the image. On second works, the disclosed AI origin counted for little in ratings of the new material, a parallel to the Office's practice of registering a person's contribution with the AI material disclaimed. It has been argued that the market's penalty on AI-labeled work, with the human-authorship rule, gives makers reasons to hide their use of AI;[26] here, disclosing an AI source cost a person's rework little in ratings, though only for students asked about that work. The creativity item did not say whose creativity, so the ratings say nothing about whether prompting alone makes a person an author. We claim relevance only in narrow senses: originality can reach a jury,[9] the line between prompting and authorship is unsettled, and a gap between a rule and the intuitions of the people it governs is worth knowing.

The remaining carry-over has no single cause. It was largest at slight edits in Study 5, mostly among people told of a 10-hour edit; it appeared at dramatic edits in Studies 2 and 5; and in Study 4 it ran the other way, not significantly (Supplementary Note 6). Every registration followed data collection. The participants were 747 students from one subject pool, 24 of them in two studies; the ten images came from one generator; nobody checked belief in the cover story; the debriefing overstated the law; effort was never rated on second works; some Slight versions were also cropped; and no second work had a source of neutral origin. The study most likely to change our conclusion would ask, about the same pairs, how creative the whole work is and how creative the person's contribution is. Others would vary editing time within each person, describe the AI as a tool rather than the maker, and ask the people who apply the rules, such as examiners and jurors.

Methods

Ethics and participants

All five studies were approved by the University of California, Los Angeles (UCLA) Institutional Review Board (protocol IRB#24-001292). Each study was an online Qualtrics survey, and participants gave informed consent by choosing 'I consent to this study' after reading the consent

form.

All participants were students in the UCLA Psychology subject pool and took part for course credit; every analyzed record came through the pool.

The studies are numbered in the order of the paper's argument, not the order in which they ran. Study 2 ran first, then Studies 4, 1 and 3, and Study 5 last. A power analysis calculates how many participants a study needs to detect an effect of a given size. None was run for any study. Each sample size came from a quota, a planned number of participants for each label group (in Study 5, for each combination of label and editing time; Supplementary Methods).

The exclusion rules were written into each registration after the data had been collected. They removed survey test runs, unfinished sessions, anyone who did not answer 'Yes, I did' when asked at the end whether they had taken part seriously and, in Studies 2 to 5, anyone who answered an arithmetic question wrongly (the Supplementary Methods give every rule). The rules left 111 of 119 exported records in Study 1, 235 of 275 in Study 2, 85 of 100 in Study 3, 117 of 140 in Study 4 and 223 of 259 in Study 5. These 771 participations came from 747 people. Of these, 24 took part twice, each in one earlier study and then in Study 5, so they had already seen some of the same originals and read the debriefing, which told them that anything produced with AI is insufficiently creative (Procedure). No registered analysis removes them, no analysis checks whether their first session changed their ratings, and the exploratory model that pools Studies 2, 4 and 5 counts twice those who had taken part in Study 2 or 4.

The mean ages in Studies 1 to 5 were 20.08, 19.96, 20.16, 20.52 and 20.22 years, and 83.8%, 81.7%, 70.6%, 82.1% and 80.3% of participants identified as female; the Supplementary Methods give the full counts.

Design and stimuli

The label told participants how the artist, Alex, had made each original image. Under the Human label he used his graphic design skills and digital sketching tools. Under the AI label he developed and input prompts into an AI system, which generated the artworks (the Supplementary Methods quote each version). The survey assigned each participant at random, each label equally often, to one label only. Both label groups saw the same images, presented in the same way, so a difference between the groups is an effect of what they were told. The Supplementary Methods give the size of each label group.

Studies 1 and 3 asked about the originals. Study 1's participants rated all ten. Study 3's rated nine, chosen at random, three at each of three stated times that Alex had spent on them: under 1 hour, 10 hours or 100 hours. Under the Human label he spent the hours drawing, and under the AI label revising prompts.

Studies 2, 4 and 5 asked about second works. Both label groups read, in the same words, that Alex had made each one from an original with his design and sketching tools. Each original was followed by three second works, each shown beside it: None, the original shown again unchanged; Slight, a light edit; and Dramatic, a heavy edit. Study 2 used all ten originals. Studies 4 and 5 used six, chosen at random, three with a human or humanoid figure and three without. Study 4 also stated how long Alex had spent on each original, which we call source time, with two originals at each of the three times. Study 5 stated these times too, but a fault

in the survey script left their record blank, so no result about source time can be reported for Study 5. Study 5 also stated how long Alex had spent making the second works, which we call editing time. Each participant was told one editing time for the whole session, assigned at random with the label, so the six combinations formed separate groups of 35 to 38 people. Stated time, source time and modification level varied within participants, each of whom saw every level.

The ten originals were made with OpenAI's DALL·E image generator, used through ChatGPT, from prompts that we wrote; no project file records their wording. Eight of the ten image files carry a provenance record, a statement of origin written into the file and digitally signed by OpenAI under the standard of the Coalition for Content Provenance and Authenticity (C2PA). Each of the eight names DALL·E and ChatGPT, marks the image as AI-generated and is dated 6 to 8 November 2024. The other two files carry no record. Five originals show a human or humanoid figure and five do not; the analyses call this the kind of image. Every image under both labels was made with AI, so the Human label was a cover story.

We made the Slight and Dramatic versions by editing each original by hand, in image-editing software that we operated manually: Adobe Photoshop and, for one Dramatic version, the online editor Dzine, used only as a manual editor, whose logo we then removed from that image. No generative or other AI editing tool changed any image. The registrations of Studies 2, 4 and 5 describe the Slight versions as color and contrast adjustments, but 7 of the ten were also cropped or resized (Fig. 1a; Supplementary Methods).

Procedure

Before every original, a short description restated the label; in Studies 3 to 5 it also stated a time (Supplementary Methods). The original then stayed on screen alone for at least 8 seconds. In Studies 2, 4 and 5, each second work was then shown beside it for at least 10 seconds. The ratings came on the next screen, without the image, in a fixed order, with a required box below them for explaining each rating. Every explanation was therefore written after its rating, although a rating could still be changed before the screen was left. The survey did not allow going back.

The originals and the second works came in random orders, not all of which were recorded, so no analysis takes order into account. No manipulation check asked whether participants had noticed, remembered or believed the label.

The final screen, a debriefing that was the same in all five studies, said: 'Current legal standards state that anything produced with artificial intelligence is insufficiently creative, even if you use AI as a base against which you create new work.' That overstates the law: the US Copyright Office registers a person's own sufficiently creative selection, arrangement or modification of AI-generated material, with that material left out of the claim.[1,2] The debriefing thus stated as law the categorical view that this paper calls the original sin and tests as a hypothesis about lay judgment, and it did not tell participants that every image had been made with AI (Supplementary Methods).

Measures

The ratings are lay judgments of creativity, effort, transformativeness and essence change. None is a judgment of authorship, of originality in the legal sense, or of whether a work can be protected. Each item offered five labeled answers, scored 1 to 5 by position. Originals were rated for creativity and for effort (the Results quote both items). Only the creativity item carried a definition, and neither says whose creativity or effort is meant. The creativity definition echoes copyright's standard of originality, at least a minimal degree of creativity (*Feist Publications, Inc. v. Rural Telephone Service Co.*, 499 U.S. 340, 345 (1991)). That standard is a floor, not a scale, so a rating of how much creativity was involved is not a judgment of whether a work clears it.

Second works were rated for transformativeness, essence change and creativity. The transformativeness item defined transformativeness as 'the degree the second work adds new expression, meaning, or message to the first work'. Studies 4 and 5 also defined essence and creativity, and Study 2 defined neither (Supplementary Methods). Effort was not rated for second works. The transformativeness definition borrows the Supreme Court's description of a transformative use in fair-use law (*Campbell v. Acuff-Rose Music, Inc.*, 510 U.S. 569, 579 (1994)). Here it is a lay rating of what the second work adds, not a fair-use judgment. In each of Studies 2, 4 and 5, two transformativeness items had a sixth answer point. As registered, their ratings (470, 140 and 274) are left out of the primary analyses.

Registrations

All five studies were registered on the Open Science Framework (OSF), an online archive for study plans, on 8 October 2026 (`https://osf.io/u3d69/overview?view_only=be795b275c3548428d74457df31a4159`), after their data had been collected, so no study was preregistered. What we had already seen of each study's data before filing, and the corrections to our earlier report of Studies 2 and 4, are set out in the Supplementary Information (Registrations).

Statistical analysis

The main quantity is the label's penalty: the mean rating under the Human label minus the mean under the AI label, in points on the 1-to-5 scale. No condition lacked a label, so a penalty cannot be split into a drop under one label and a rise under the other.

The registered analyses fitted a linear mixed model to every individual rating of each question, leaving missing ratings out (the Supplementary Methods give the counts). Each model gave every participant and every original its own baseline, called a random intercept. A participant or an original rated high throughout therefore did not distort comparisons between conditions. Each model included the factors that its study varied and all their interactions, an interaction being a change in one factor's effect across the levels of another. The kind of image was crossed with those factors in Studies 1 and 2, added without interactions in Study 5 and left out in Studies 3 and 4 (the Supplementary Methods give each model). The models did not let the label's effect differ from one original to the next. Their intervals therefore describe these ten images and do not allow for how the penalty might vary across other images.

Each reported difference compares the model's predicted mean ratings for two conditions, each averaged with equal weight over the other factors. Tests were two-sided, counting a difference in either direction, at .05. Their degrees of freedom, which set how large a statistic must be to count as significant, came from Satterthwaite's approximation. Study 1's contrasts used the Kenward–Roger approximation, which gives the same degrees of freedom for its penalty (Supplementary Methods).

Within each study, the tests that its registration names as confirmatory were corrected together by Holm's method.[21] The corrected sets held 4, 12, 6, 15 and 6 tests in Studies 1 to 5. Study 1's four are H1 on each rating, H2 and H4. H4, whether the AI label spreads ratings more between participants and less within each participant, is decided by two ratios of variances, a measure of spread (Supplementary Methods). It enters the corrected set as a likelihood-ratio test of whether letting the variance between participants differ by label improves the model's fit. Study 1's H3, whether the penalty is the same for originals with and without a figure, is decided by its own registered rule outside the set. So are Study 3's confirmatory equivalence tests and decision rule.

Where a registration asked whether an effect was small, it used an equivalence test, two one-sided tests at .05 that pass when the effect's 90% confidence interval lies wholly inside a stated bound.[22,23] In Studies 2 and 4, the larger one-sided p-value also had to survive Holm's correction.

Three bounds were used, in rating points. Study 1's H3 used ± 0.20, a fifth of a scale point, set after its analyses had been run. Studies 2, 4 and 5 used ± 0.224 for the label's effect on all three second-work measures, and Study 3 used it for the label's gap at 100 hours. It is half of Study 1's registered creativity penalty of 0.448 points. The bound is a fixed number, although the penalty it halves was itself estimated with uncertainty (95% CI 0.19 to 0.70). Each registration set it knowing Study 1's estimate and, for Studies 2 to 4, the earlier analyses of its own data. Study 5 tested editing time against ± 0.219, Study 3's registered creativity contrast between 100 hours and under 1 hour, by a rule fixed before any Study 5 model was fitted (Supplementary Methods).

Model checks and further registered analyses

As registered checks, the confirmatory models were refitted as ordinal models (cumulative-link mixed models), which treat the answers as ordered categories. In Study 1, agreement of this ordinal model was part of the registered rule for H2. Elsewhere a disagreement is reported beside the verdict rather than replacing it. A model that did not converge yields no p-value, interval or verdict, and its check is reported as not completed; every study's code applies this rule. A singular fit keeps its numbers and is reported as singular. Supplementary Note 2 and the Supplementary Methods describe the other registered analyses, among them checks in models that let effects vary between participants and between originals, Study 1's robustness checks, refits of Studies 1 and 4 without the Human-label participants who wrote that a work was machine-made, and Study 5's rescaled reanalysis.

The written explanations

Eight fixed word lists, sets of words and phrases that a script searches for in every explanation in the same way, flagged what each explanation mentioned, such as visible detail, time, skill, prompting or machines, or a statement that a machine made the work (the Supplementary Methods list all eight). A flag records that a topic was mentioned, not that it was endorsed. We wrote the lists by reading Study 1's explanations with the label and the rating hidden. Four parts of them, among them the whole list for a statement that a machine made the work, were settled after results by label had been read. Studies 2 to 5 ran the lists unchanged. The script received only identifiers and text. No person and no language model labeled, coded or judged any explanation for these measures. An earlier language-model coding of Study 1's explanations, made with the label in its instructions, was retired before registration and is not used.

Each registration names two registered secondary checks on the explanations. All other analyses of them are exploratory, among them Study 1's five tests corrected by the Benjamini–Hochberg method,[27] whose results are descriptive readings, not verdicts (Supplementary Methods). Because each explanation was written after its rating, none of these measures can show why anyone rated as they did.

Exploratory analyses

The exploratory analyses were specified on 9 and 10 October 2026, after every registered result was known. Their tests are not corrected as a set, and none changes a registered verdict. The Supplementary Methods and Supplementary Notes 4 to 7 describe them.

Software and reproducibility

The models ran in R 4.5.2 (lme4 2.0-1, lmerTest 3.2-1, emmeans 2.0.3, ordinal 2025.12-29 and brms 2.23.0), and the text measures and the image check in Python 3.12.3. The Supplementary Methods give the other versions, those of the pretrained models and the fixed starting values of the random-number generators. Each analysis records a checksum, a fingerprint of a file's exact bytes, for every input and output, and a script reads every result, and every count of participants, records or ratings, from the analysis tables and records the table, row and column behind each.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The de-identified survey data of all five studies, the five survey files, the 30 stimulus images and the six registration documents are available at `https://anonymous.4open.science/r/AICopyrightabilityVisualArt-2B38/`, an anonymized copy of the lab's repository. De-identification emptied each record's IP address and location and replaced the participant number with a code,

the same code for a person in every study; the ratings, the written explanations and every other answer are unchanged. Source data are provided with this paper: every value plotted in the figures, with the analysis table, row and column it comes from, is in `Docs/Writing/Final Paper/figures/plotted_values.csv` in the same repository. The registrations are on the Open Science Framework at `https://osf.io/u3d69/overview?view_only=be795b275c3548428d74457df31a4159`. The two published word-norm datasets used in exploratory analyses are not redistributed and are available from their authors.[28,29]

Code availability

The analysis code is available at `https://anonymous.4open.science/r/AICopyrightabilityVisualArt-2B38/`. It holds the R and Python notebooks for each study and for the cross-study analyses, the shared text module that holds the eight word lists, and the image check, with the tables and figures that the code wrote and run records that give each file's checksum, a fingerprint of its exact bytes. It also holds the manuscript's sources and the scripts that write every number, table and figure in this paper from those tables, recording the table, row and column behind each number and plotted value. The repository keeps the project's folder layout, so every path in the code works as it stands, and its README gives the order in which to run the notebooks. Three pretrained models come from Hugging Face, a site that hosts trained models: all-mpnet-base-v2, DINOv2 and CLIP. The image check fetches the two image networks at fixed versions; the text code loads the sentence-embedding model from a copy on the computer, revision e8c3b32 here.

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Author contributions

D.G.K. conceived and designed the studies, collected and analyzed the data, wrote the analysis code and wrote the paper. A.L. contributed to the conception of the studies and collected the data. K.J.H. contributed to the conception of the studies and to writing and editing the paper.

Competing interests

The authors declare no competing interests.